

VISUALIZATION

Visual Encoding

Visual Language is a Sign System

- Image perceived as a set of signs
- Sender encodes information in signs
- Receiver decodes information from signs



Semiology of Graphics [J. Bertin, 83]

Marks & Channels

- **Marks:** represent **items** or **links**
- **Channel = Visual Variable**
- **Channels:** change **appearance** based on **attribute**

Marks for Items

Basic geometric elements

➞ Points



0D

➞ Lines



1D

➞ Areas



2D

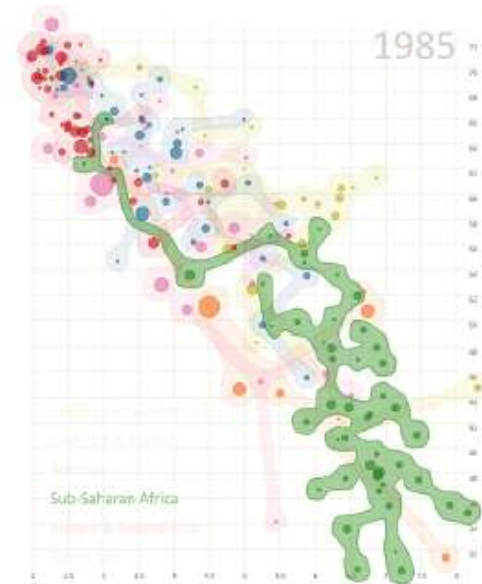
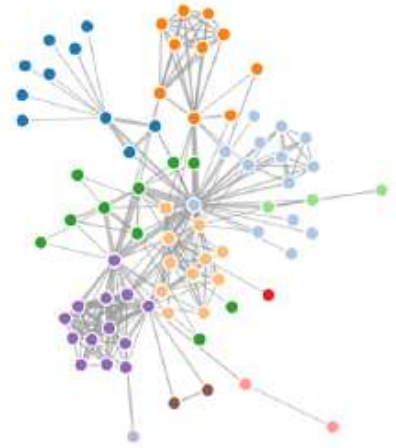
3D mark: Volume, but rarely used

Marks for Links

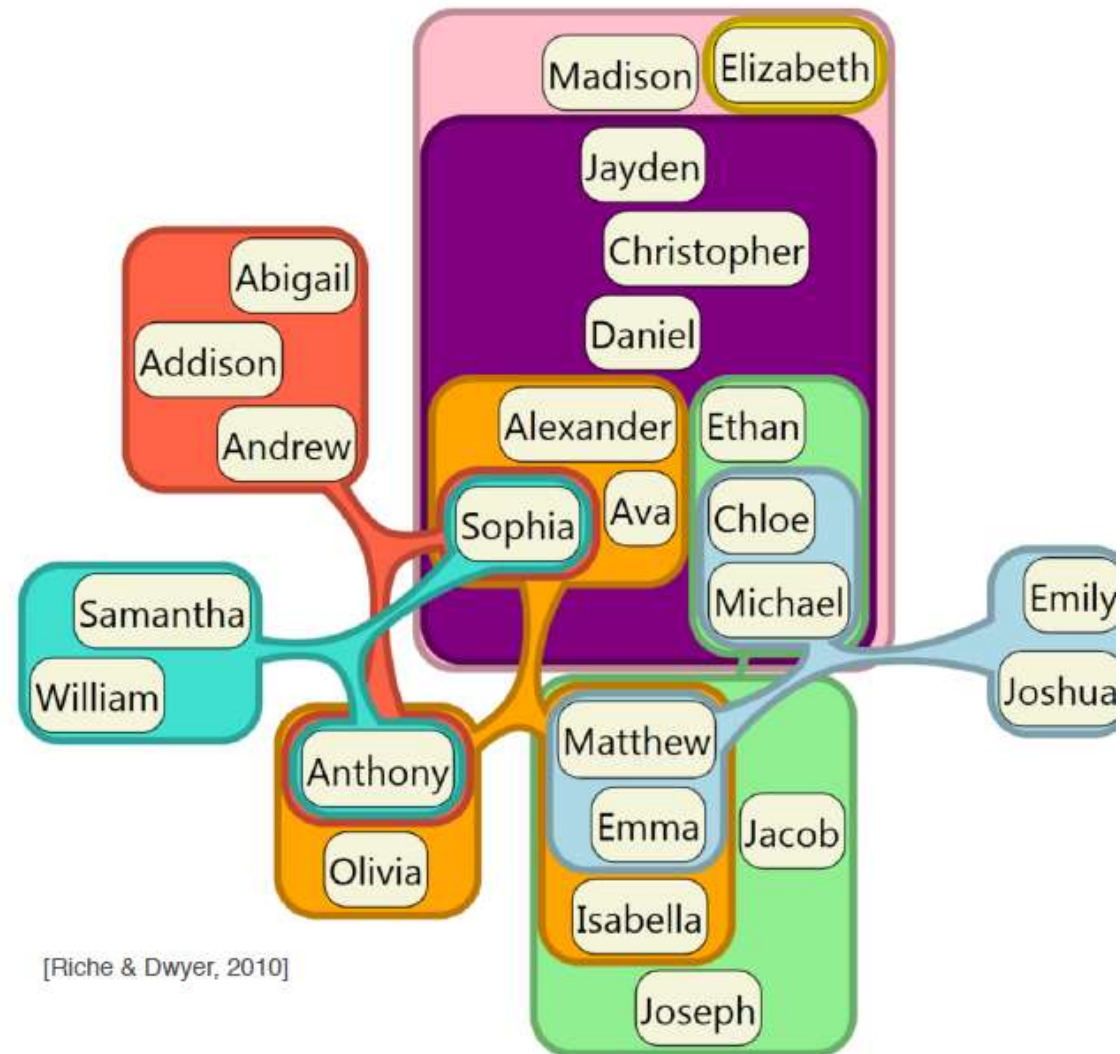
➔ Containment



➔ Connection



Containment can be nested



[Riche & Dwyer, 2010]

Channels (aka Visual Variables)

Control appearance
proportional to or
based on attributes

→ Position

→ Horizontal



→ Vertical



→ Both



→ Color



→ Shape



→ Tilt



→ Size

→ Length



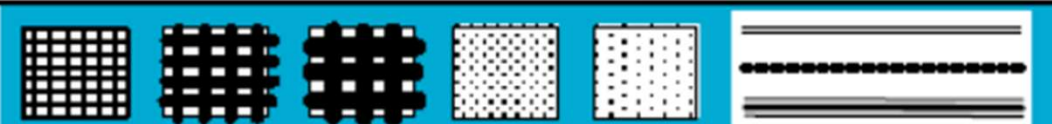
→ Area



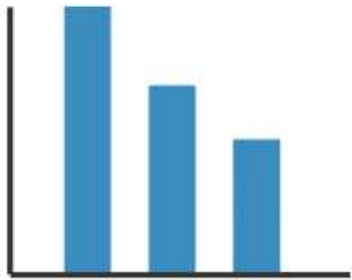
→ Volume



Bertin's Visual Variables

Bertin's Original Visual Variables	
Position changes in the x, y location	
Size change in length, area or repetition	
Shape infinite number of shapes	
Value changes from light to dark	
Colour changes in hue at a given value	
Orientation changes in alignment	
Texture variation in 'grain'	

Using Marks and Channels

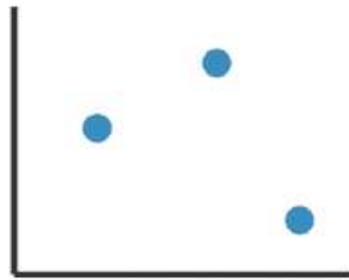


Mark: Line

Channel: Length/Position

1 quantitative attribute

1 categorical attribute



Mark: Point

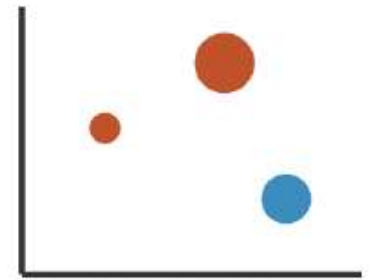
Channel: Position

2 quantitative attr.



Adding Hue

+1 categorical attr.



Adding Size

+1 quantitative attr.

Mapping Visual Variables to Attribute Types

property	marks	ordinal/nominal mapping	quantitative mapping
shape	glyph	○ □ + △ S U	
size	rectangle, circle, glyph, text		
orientation	rectangle, line, text		
color	rectangle, circle, line, glyph, y-bar, x-bar, text, gantt bar		

Types of Channels

Magnitude Channels

How much?

Position

Length

Saturation ...

Identity Channels

What? Where?

Shape

Color (hue)

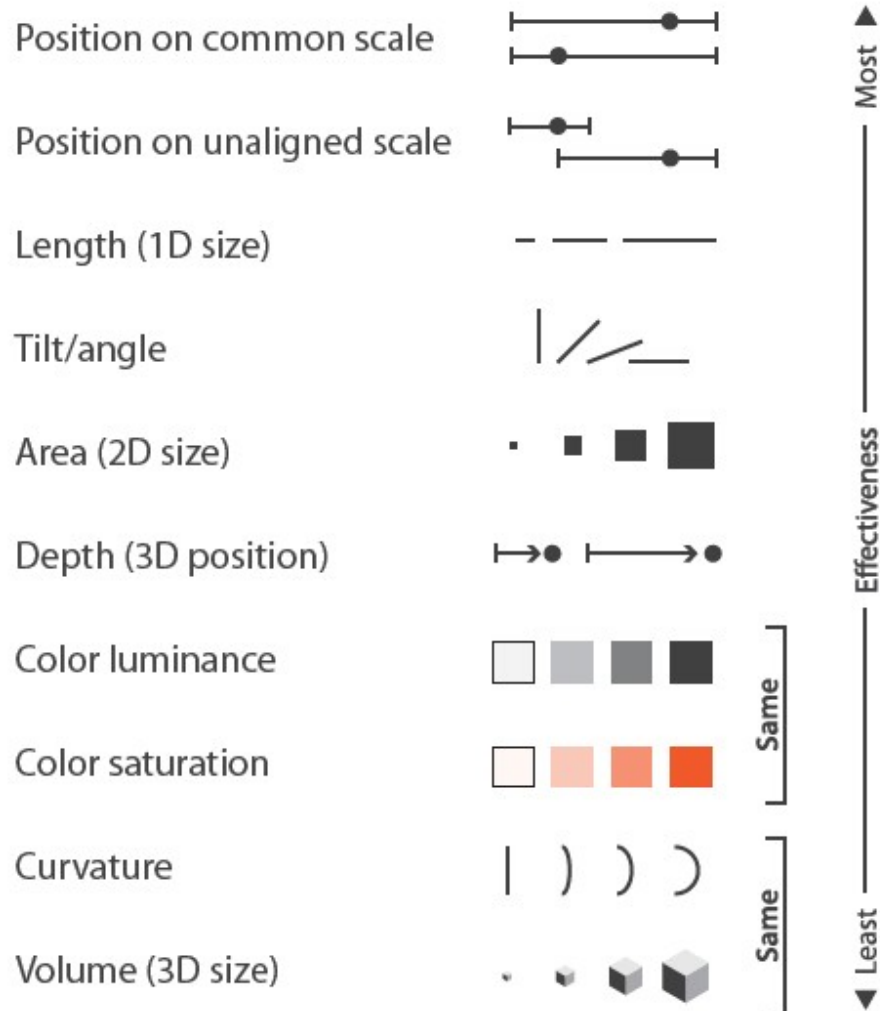
Spatial region ...

Ordinal & Quantitative Data

Categorical Data

Channels: Expressiveness Types and Effectiveness Ranks

➔ Magnitude Channels: Ordered Attributes



➔ Identity Channels: Categorical Attributes



Characteristics of Channels

Selective

Is a mark distinct from other marks?

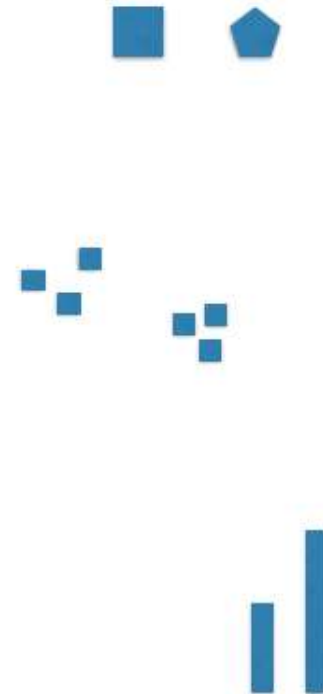
Can we make out the difference between two marks?

Associative

Does it support grouping?

Quantitative (Magnitude vs Identity Channels)

Can we quantify the difference between two marks?



Characteristics of Channels

Order (Magnitude vs Identity)

Can we see a change in order?



Length

How many unique marks can we make?

Position

Strongest visual variable

Suitable for all data types

Problems:

- Sometimes not available
(spatial data)

- Cluttering



Selective: yes

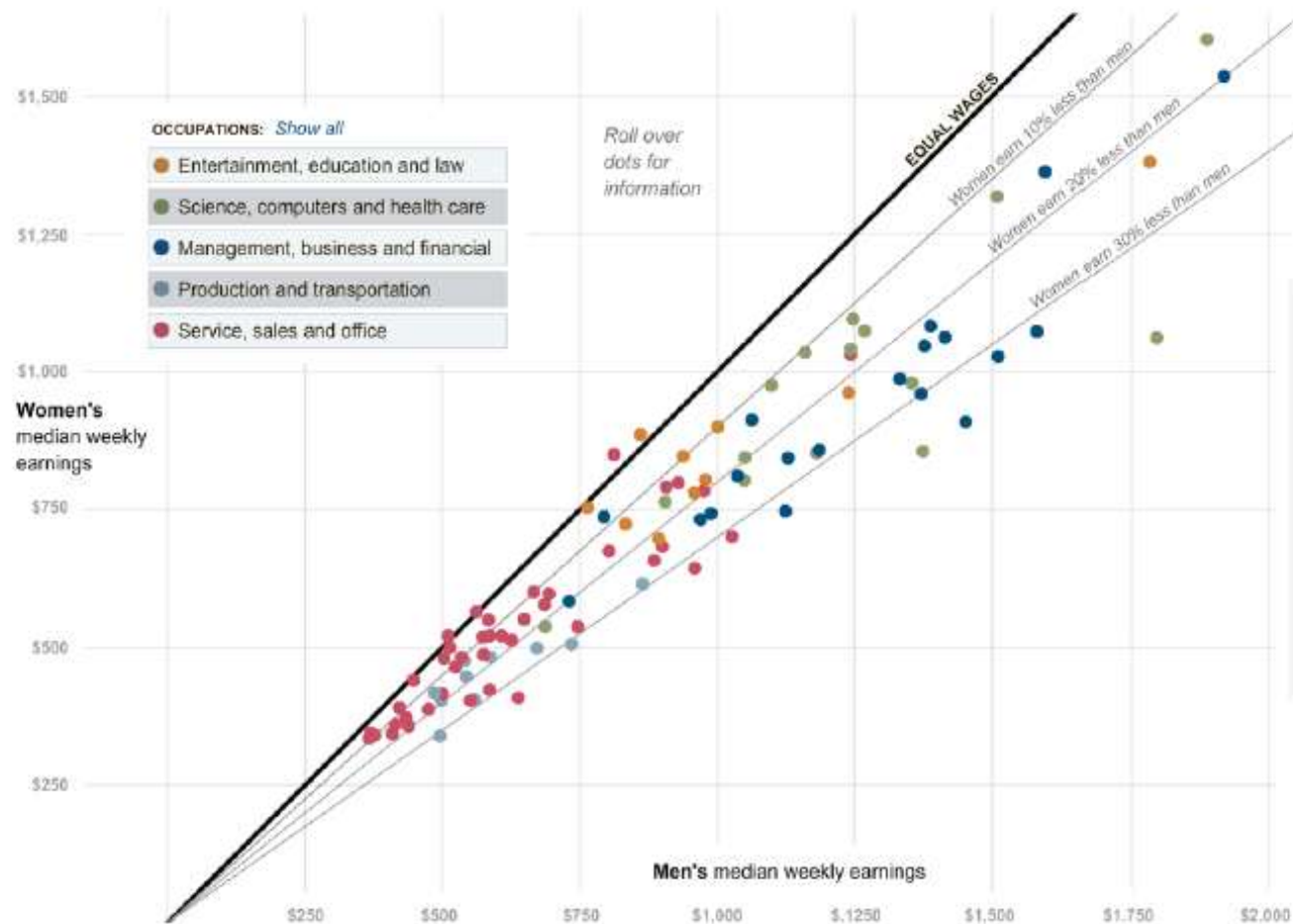
Associative: yes

Quantitative: yes

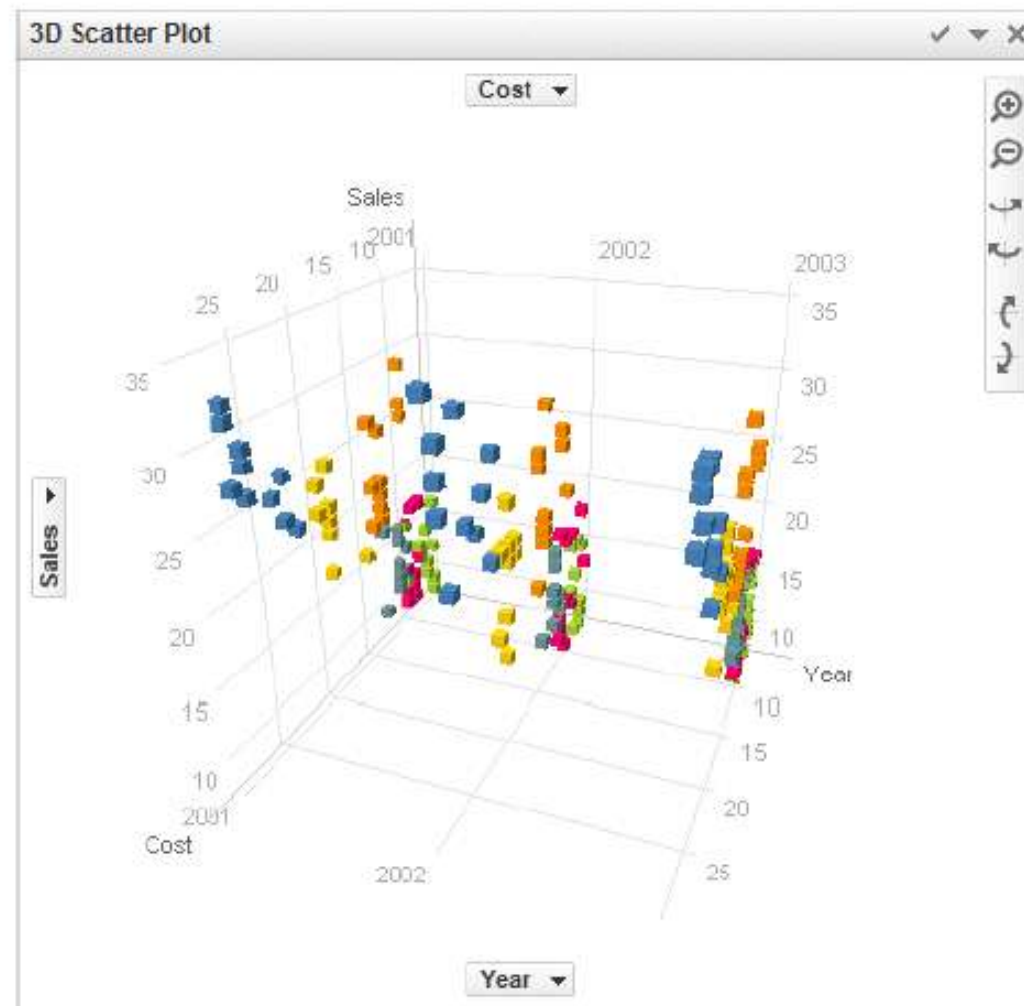
Order: yes

Length: fairly big

Example: Scatterplot



Position in 3D?



[Spotfire]

Length & Size

Good for 1D, OK for 2D, Bad for 3D

Easy to see whether one is bigger

Aligned bars use position redundantly

For 1D length:

Selective: yes

Associative: yes

Quantitative: yes

Order: yes

Length: high



Example 2D Size: Bubbles

Four Ways to Slice Obama's 2013 Budget Proposal

Explore every nook and cranny of President Obama's federal budget proposal.



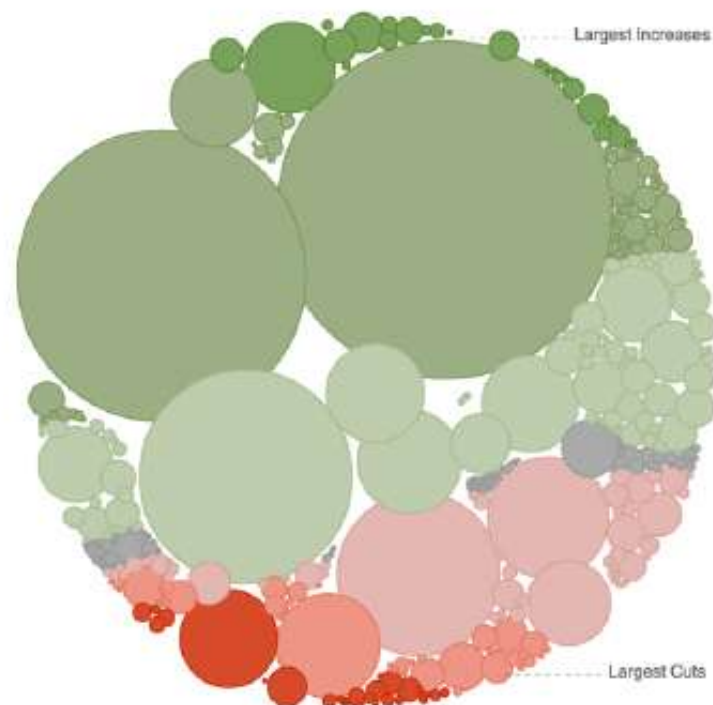
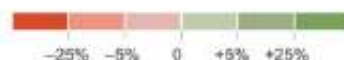
How \$3.7 Trillion Is Spent

Mr. Obama's budget proposal includes \$3.7 trillion in spending in 2013, and forecasts a \$901 billion deficit.

Circles are sized according to the proposed spending.



Color shows amount of cut or increase from 2012.



Color

- Color is an important component of any visualization.
- Color can be based on the RGB model.
- Color HSV is a common cylindrical-coordinate representation of points in an RGB color model.

Color HSV

- Based on polar coordinates, not Cartesian coordinates.
- HSV is a non-linearly transformed (skewed) version of RGB cube
 - Hue: quantity that distinguishes color family, say red from yellow, green from blue
 - Saturation (Chroma): color intensity (strong to weak). Intensity of distinctive hue, or degree of colour sensation from that of white or grey
 - Value (luminance): light color or dark color
- Each of these components can be utilized as visual variables.
- Note that hue is equivalent to the Color of Bertin's visual variables.
- Similarly, value is related to the Value of Bertin's visual variables.

Information in Color

- (Grey)Value is perceived as ordered (O)



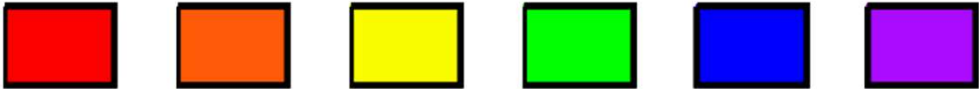
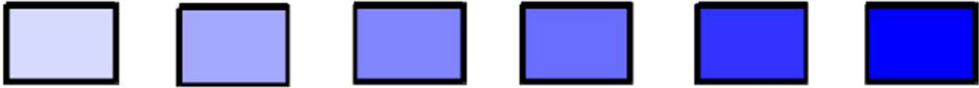
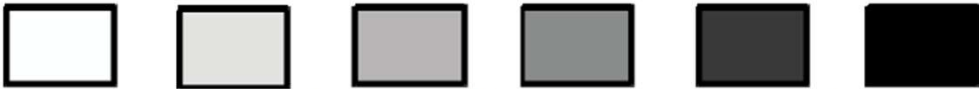
- Can encode quantitative values (Q) [not as well]



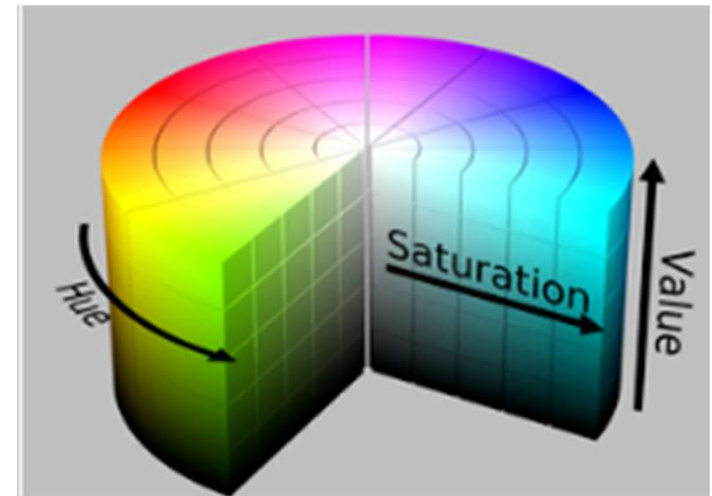
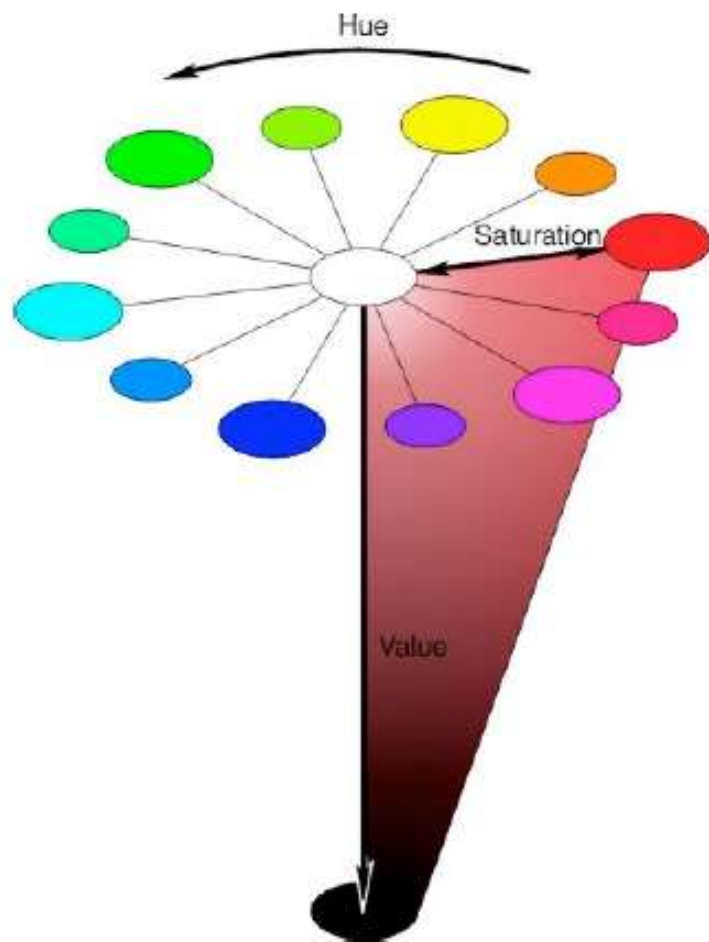
- Hue is normally perceived as unordered (N)



Color HSV

- Hue 
 - Wavelength (red, green, yellow, blue)
 - Spectrum (VIBGYOR)
- Saturation 
 - Pastel versus strong (baby blue, sky blue, royal blue)
- Value 
 - amount of energy (white, light gray, dark gray, black)
 - Usually $V = 0.299*R + 0.587*G + 0.114*B$

HSV



Value/Luminance/Saturation

OK for quantitative data when length & size are used.

Not very many shades recognizable

Selective: yes

Associative: yes

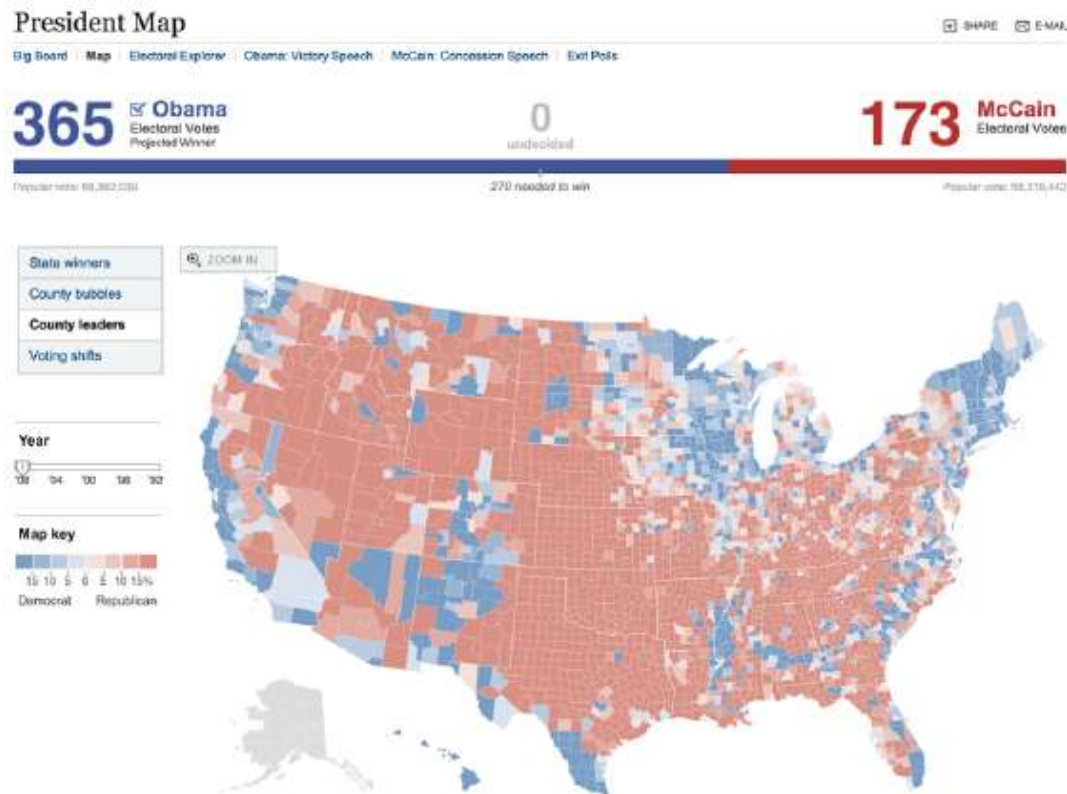
Quantitative: somewhat (with problems)

Order: yes

Length: limited



Example: Diverging Value-Scale

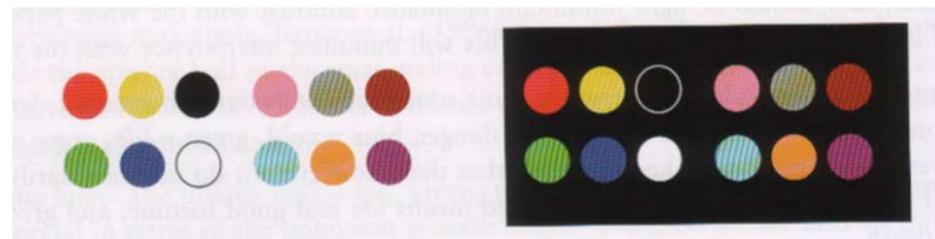


Color Hue

- Good for qualitative data (identity channel)
- Limited number of classes/length (~7-10!)
- Does not work for quantitative data!
 - Selective: yes
 - Associative: yes
 - Quantitative: no
 - Order: no
 - Length: limited

Color for Categories

- Can different colors be used for categorical variables?
 - Yes (with care)
 - Ware's suggestion: 12 colors
red, green, yellow, blue, black, white, pink, cyan,
gray, orange, brown, purple



Color for Sequences

Can you order these (low->hi)



Possible Color Sequences

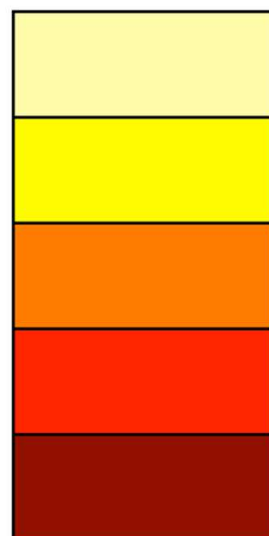
Gray scale



Full spectral scale



Single sequence
part spectral scale



Single sequence
single hue scale



Double-ended
multiple hue scale



Color: Bad Example

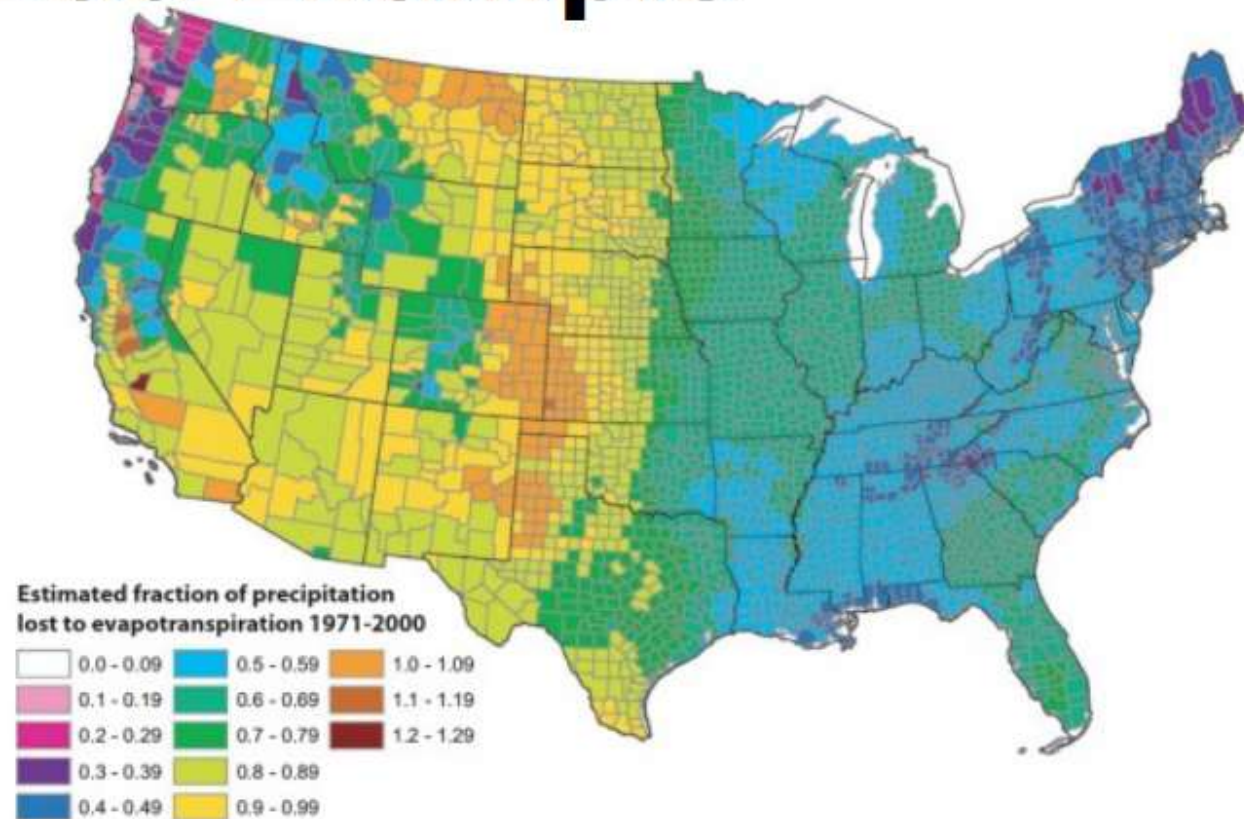


FIGURE 13. Estimated Mean Annual Ratio of Actual Evapotranspiration (ET) to Precipitation (P) for the Conterminous U.S. for the Period 1971-2000. Estimates are based on the regression equation in Table 1 that includes land cover. Calculations of ET/P were made first at the 800-m resolution of the PRISM climate data. The mean values for the counties (shown) were then calculated by averaging the 800-m values within each county. Areas with fractions >1 are agricultural counties that either import surface water or mine deep groundwater.

Cliff Mass

Shape

- Great to recognize many classes.
 - No grouping, ordering.
 - Selective: yes
 - Associative: limited
 - Quantitative: no
 - Order: no
 - Length: vast

Icons



Mapping to Data Types

	Nominal	Ordinal	Quantitative
Position	✓	✓	✓
Size	✓	✓	✓
(Grey)Value	✓	✓	~
Texture	✓	~	✗
Color	✓	✗	✗
Orientation	✓	✗	✗
Shape	✓	✗	✗

✓ = Good

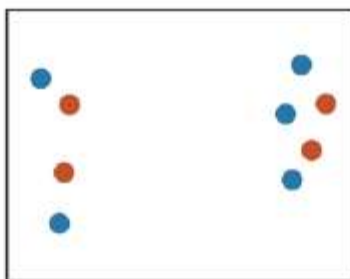
~ = OK

✗ = Bad

Separability of Attributes

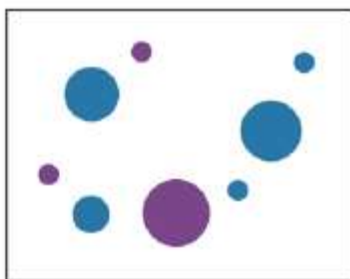
Can we combine multiple visual variables?

Position
+ Hue (Color)



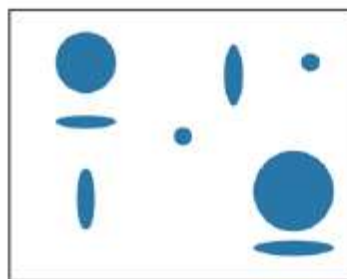
Fully separable

Size
+ Hue (Color)



Some interference

Width
+ Height



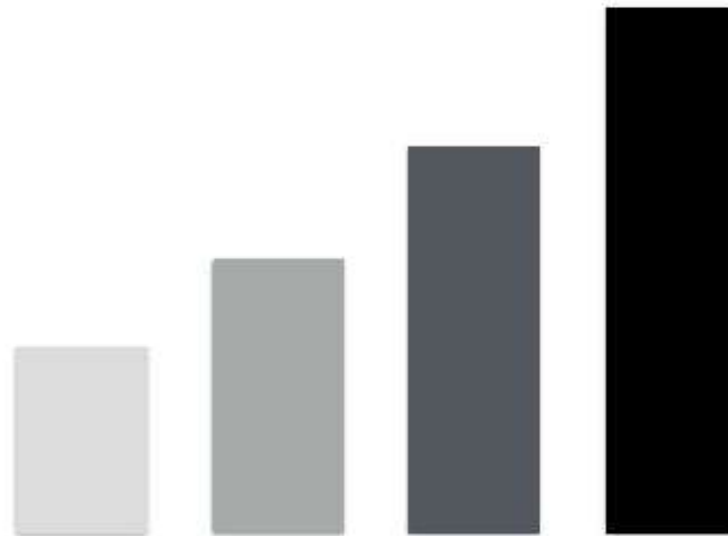
Some/significant
interference

Red
+ Green



Major interference

Redundant encoding



Length, Position and Value

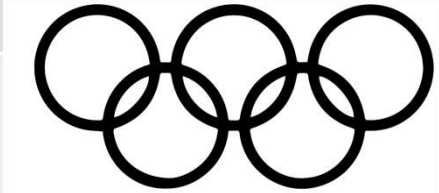
Gestalt Principles of Pattern Perception

- Gestalt Principles are principles/laws of human perception that describe how humans group similar elements, recognize patterns and simplify complex images when we perceive objects.
- Gestalt is German for unified whole.
- German psychologists Max Wertheimer, Kurt Koffka, and Wolfgang Kohler created the Gestalt Principles in the 1920s.
- Information Visualization should use these principles to design effective visualizations.



Max Wertheimer
1880-1943

Gestalt Law of Pattern Perception (1/2)

Proximity	Things close together are perceptually grouped together, separating them from those farther apart.	
Closure	We try to see collections of objects as creating a larger, more complete objects	
Similarity	We group similar objects together	 
Simplicity/ Prägnanz (German for <i>good figure</i>)	Brain will perceive ambiguous shapes in as simple a manner as possible.	

Gestalt Law of Pattern Perception (2/2)

Continuity

We group elements that seem to follow a continuous path in a particular direction.

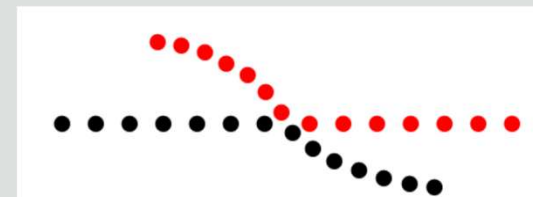
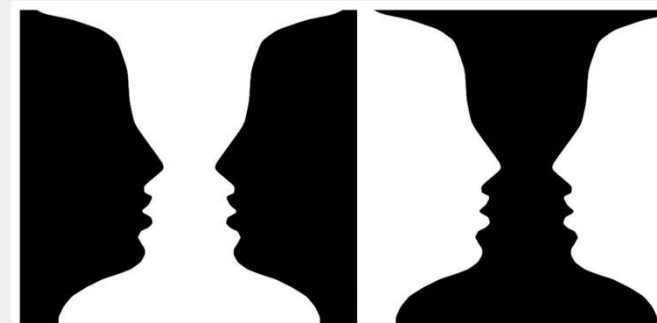


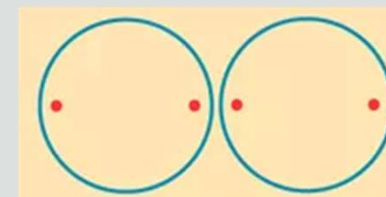
Figure & Ground

Brain will distinguish between the objects it considers to be in the foreground of an image (the figure, or focal point) and the background (the area on which the figures rest).



Common region

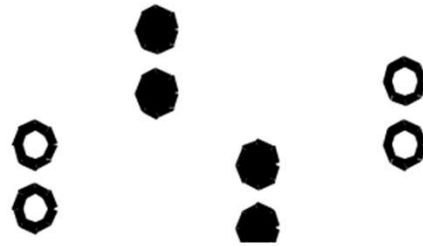
We perceive elements that are in the same closed region as one group.



Combining Gestalt Principles

- Several Principles can be combined

- Proximity reinforces similarity



- Proximity reinforces closure/symmetry



- Proximity opposes closure/symmetry

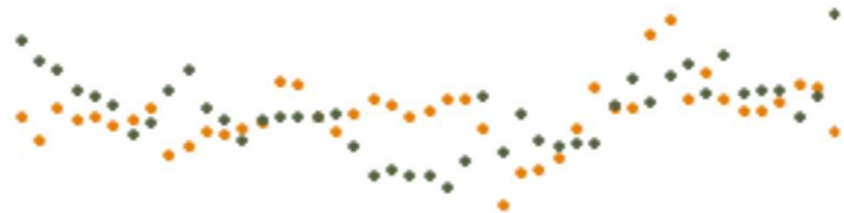


Combining Similarity & Connectedness

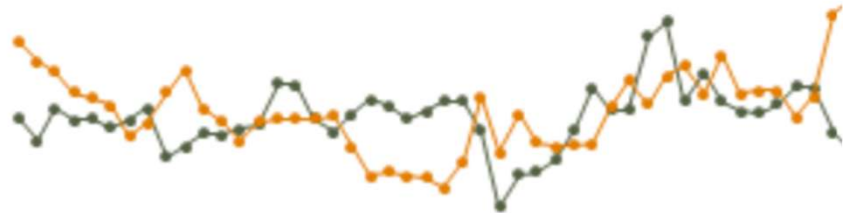
A bunch of dots:
no visual structure



Colored dots:
visual structure emerges



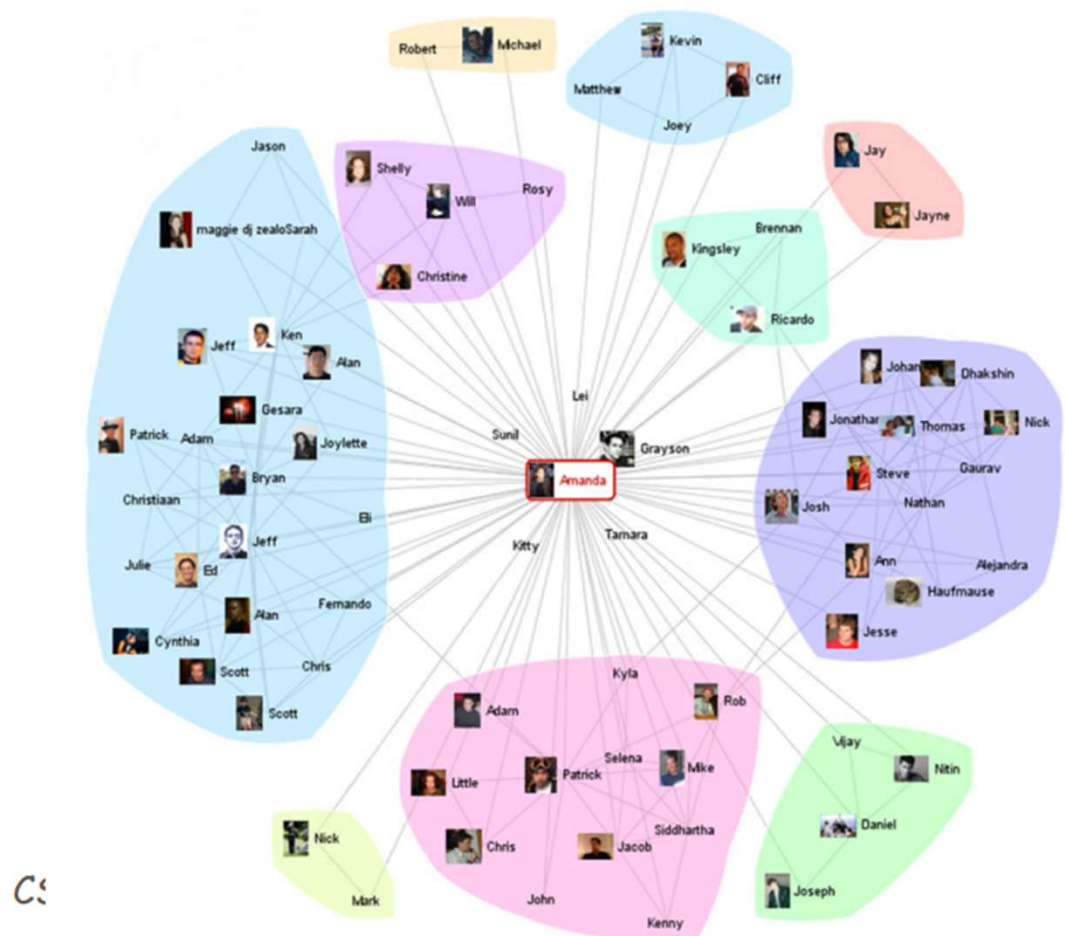
Colored & connected dots:
strong visual structure



Combining Similarity & Proximity

Grouping created by

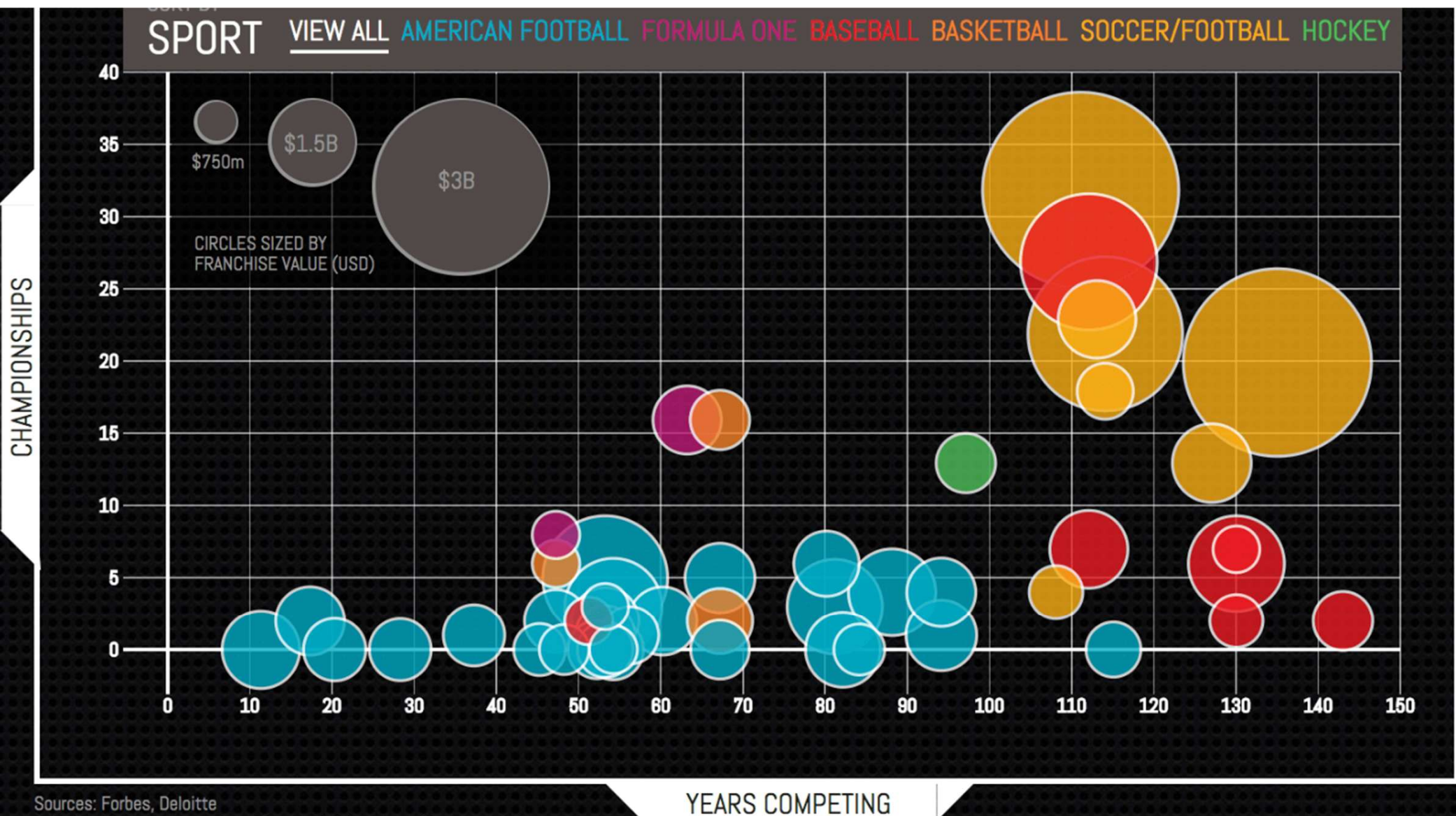
- Proximity
- Similar background



Exercise

- For the given visualizations
 - Type of visualization
 - Determine the data that is represented
 - Semantics
 - Data Item
 - Data Attributes (including type)
 - Dimension
 - Dataset Type
 - Determine Visual Encoding of the data
 - Any criticism of the visualization?

Most Valuable Sports Franchises



Reading

- Chapter 3